

NETRA — Offline-first Statutory Compliance Auditor

Model Round 1 (netra_roi / YOLO26n) — Verification Report

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1. Model provenance

Field	Value
Base model	yolo26n.pt (Ultralytics) · 2.5M params · 5.3 GFLOPs
Classes (model.names order)	PACKAGE, PDP, PRICE, BARCODE, BOP
Metrics (synthetic corpus)	mAP50 0.995 · mAP50-95 0.953 · PRICE weakest 0.859 (ROI hint only)
sha256 netra_roi.pt	271aafe0e34dca31f3aa2cfa75d717e2cc8fc808b00da22a14c6f61e79ee9eab5
sha256 yolo26n_roi.tflite	a581ff30c840b3ea4962b7bc584150e9b0499d369af42753102af958da6dec42
License note	Ultralytics AGPL-3.0; research prototype for SIH evaluation, not distributed. Production path: RF-DETR (Ap

2. LiteRT export contract (empirically pinned)

Property	Pinned value
Input tensor	640×640×3, NCHW float32, caller divides by 255 (graph_normalizes_input=false)
Output tensor	1×9×8400 native — rows 0-3 cx,cy,w,h normalized [0,1] (normalized_0_1); rows 4-8 post-sigmoid scores
NMS / conf gate	No embedded NMS in LiteRT export — owned by NetraYolo.kt (Kotlin) and identical Python mirror
Conf / IoU thresholds	0.25 / 0.45 (served via vision_config.json — single source of law)

3. Export fidelity & golden parity

Measurement	Result
Max box deviation, .tflite vs .pt on identical tensor	0.0152 px
Max score deviation	6.20e-06
Golden detections reproduced by decoder (Python + Kotlin mirror)	100%
Classes covered by golden fixture	PACKAGE, PDP, PRICE
Fixture tamper evidence	input tensor sha256 + recorded reference enforced in CI (test_yolo_golden.py, stdlib-only, runs

4. Verification gates

Python suite: 350 passed, 3 skipped (353 collected)

Golden contract tests: 5/5 — fixture integrity, tensor contract, decoder equivalence, recorded export parity, vision_config ↔ golden single-source-of-law.

CI workflows: python (3.11/3.13, Tesseract-equipped) and android-compile (Gradle 9.3.1 + Chaquopy, Kotlin compile gate). Run history: <https://github.com/varun2507027108-oss/NETRA/actions>

On-device pipeline (pre-YOLO baseline): 0.45 ms statutory core (s5+s6, 25-run mean), 82 ms full scan-with-dossier, ~2 ms verdict. Re-measurement with ROI detection in the loop: pending.

5. Pending at time of report

Real-world accuracy (golden_report.json): NOT YET RECORDED — three real package photos pending; this report will be regenerated when they land. Synthetic-corpus mAP is a training metric, not a field claim.

Outstanding: scan_tokens v1.4.0 ROI wiring (model currently landed but not yet invoked in the scan path), s2 ROI hook, device-side golden parity run, Step 3 screens, Phase E rehearsal.